

Learning From Synthetic CT Images via Test-Time Training for Liver Tumor Segmentation

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Abstract—Automatic liver tumor segmentation could offer assistance to radiologists in liver tumor diagnosis, and its performance has been significantly improved by recent deep learning based methods. These methods rely on large-scale well-annotated training datasets, but collecting such datasets is time-consuming and labor-intensive, which could hinder their performance in practical situations. Learning from synthetic data is an encouraging solution to address this problem. In our task, synthetic tumors can be injected to healthy images to form training pairs. However, directly applying the model trained using the synthetic tumor images on real test images performs poorly due to the domain shift problem. In this paper, we propose a novel approach, namely Synthetic-to-Real Test-Time Training (SR-TTT), to reduce the domain gap between synthetic training images and real test images. Specifically, we add a self-supervised auxiliary task, *i.e.*, two-step reconstruction, which takes the output of the main segmentation task as its input to build an explicit connection between these two tasks. Moreover, we design a scheduled mixture strategy to avoid error accumulation and bias explosion in the training process. During test time, we adapt the segmentation model to each test image with self-supervision from the auxiliary task so as to improve the inference performance. The proposed method is extensively evaluated on two public datasets for liver tumor segmentation. The experimental results demonstrate that our proposed SR-TTT can effectively mitigate the synthetic-to-real domain shift problem in the liver tumor segmentation task, and is superior to existing state-of-the-art approaches.

Index Terms—Liver tumor segmentation, synthetic images, test-time training.

I. INTRODUCTION

LIVER cancer is one of the most deadly malignant diseases worldwide, accounting for over 2 million deaths every

year [1], [2]. Computed tomography (CT) imaging remains the most utilized modality for liver-related imaging tests, and liver tumor segmentation from CT volumes is a prerequisite for many clinical applications. Deep learning based methods have obtained impressive progress for automatic liver tumor segmentation [3]–[5]. These leading approaches rely on large-scale well-annotated datasets, whereas collecting such datasets is time-consuming and labor-intensive, which could hinder their performance in practical situations. Recently, learning from synthetic data is an encouraging solution to address this problem. In the context of tumor segmentation, synthetic tumors can be injected to healthy images to form training pairs, such that only image-level labels are required rather than pixel-level labels. In this paper, we train the deep neural network for liver tumor segmentation using synthetic tumor images, which can significantly reduce manual annotation efforts.

With the development of computer graphic algorithms, realistic synthetic data has been applied in many computer vision tasks [6]–[9]. Recently, medical image synthesis is also becoming an active research topic. Approaches based on Generative adversarial network (GAN) have been used for same-domain [10]–[12] and cross-domain image synthesis [13]–[15]. Moreover, methods based on GAN and morphological operations are proposed to generate synthetic tumors [16]–[19], which can be used to train the tumor segmentation network. Even though synthetic data provides many advantages, there are still differences between synthetic and real tumors. The segmentation performance shows an obvious degradation when directly applying the model trained using the synthetic images on the real test images. The domain shift problem motivates us to investigate a decent solution so as not to degrade the practical value of the synthetic data.

Unsupervised domain adaptation (UDA) is one of the promising approaches for tackling the domain shift problem, and has achieved successful performance in many tasks [20]–[28]. However, most UDA approaches require that source domain data and target domain data should be available at the same time, which may not always be feasible in real-world applications. An alternative approach is to directly adapt the model on the test sample from the target domain [29]–[32]. Test-Time Training (TTT) is one of the representative methods, and our work falls in this line of research. The concept of TTT is first introduced in [29], which proposes to add a self-supervised auxiliary task sharing some parameters with the main task. The auxiliary task can be used to update model parameters during test-time since it does not require manual labels. Ideally, the updated shared parameters can also improve

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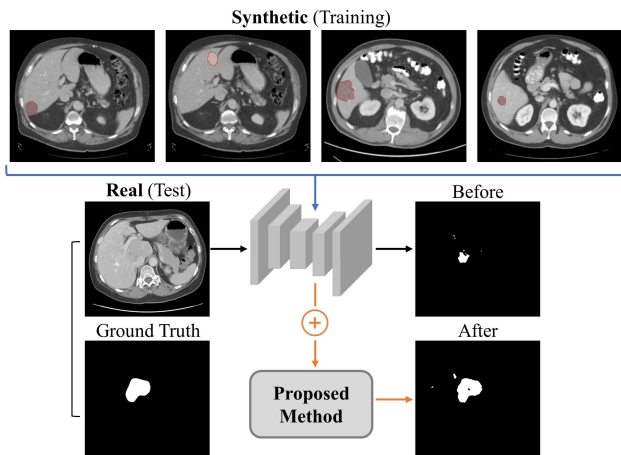


Fig. 1. Directly applying the model trained using the synthetic images on the real test image performs poorly due to the domain shift problem, and our method is proposed to tackle this problem.

the performance of main task. Even though the empirical result is encouraging in the original TTT method, its success is built upon a strong assumption that the auxiliary task and the main task should have correlated gradients between their loss functions, which is difficult to achieve. Moreover, the majority of model parameters from the main task are frozen during test-time such that they might not be able to provide sufficient flexibility to obtain satisfactory results.

In this paper, we introduce a novel Synthetic-to-Real Test-Time Training (SR-TTT) method to bridge the domain gap between synthetic training data and real test data. To the best of our knowledge, this is the first work in the literature to propose test-time training for tackling the synthetic-to-real domain shift problem in medical image segmentation. Unlike the original self-supervised TTT method which relies on the implicit connection by sharing some model parameters, SR-TTT attempts to build an explicit connection by taking the output of the main task as the input of the auxiliary task. In this way, training with self-supervision from the auxiliary task will directly affect the output of the main task. The comparison of our proposed SR-TTT with the original self-supervised TTT is shown in Fig. 2. Inspired by [33], we propose to leverage two-step reconstruction as the auxiliary task. The main task output (tumor mask) is taken as an input and fed to the auxiliary task. The reconstruction loss of the auxiliary task can be used to update the parameters of the segmentation network without requiring any manual labeling during test-time, so that it can be used to fine-tune the segmentation model on each test image. Our method updates all parameters of the segmentation model, allowing it to work at full capacity. Moreover, two-step reconstruction can be seen as a serialization task with two steps, which is hard to train. Inspired by the training techniques used in the recurrent neural network (RNN) which has great advantages in serialization tasks [34], we design a scheduled mixture strategy for training the model, in order to avoid error accumulation and stabilize the training process.

The main contributions are summarized as follows:

- We introduce a novel approach, namely Synthetic-to-Real Test-Time Training (SR-TTT), for learning from

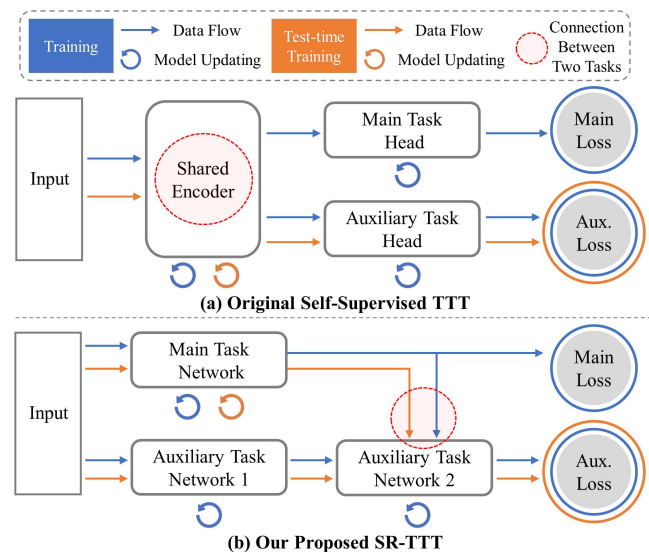


Fig. 2. Comparison between (a) original self-supervised TTT, and (b) our proposed SR-TTT. Original TTT relies on the implicit connection by sharing some parameters, while SR-TTT builds an explicit connection by taking the output of the main task as the input of the auxiliary task.

synthetic images in the liver tumor segmentation task, which can effectively reduce the domain gap between synthetic training images and real test images.

- We propose to adapt the liver tumor segmentation model trained using synthetic images to each test sample with self-supervision from an auxiliary two-step reconstruction task, and design a scheduled mixture strategy to reduce the model training difficulty.
- We conduct experiments on two liver tumor segmentation datasets, and the experiment results show that our method consistently provides performance improvement, demonstrating the effectiveness of the proposed method.

II. RELATED WORK

A. Learning From Synthetic Images

Learning from synthetic images has been extensively studied in the computer vision community for various tasks [6]–[9]. Compared to real data, synthetic data is perfectly labeled and fully controllable, which can reduce the burden of manual labeling. Recently, synthetic data is also gaining attention in medical image analysis tasks. Unlike natural images, most computer graphic algorithms are not suitable for medical images and the representative approaches for medical image synthesis are based on Generative adversarial network (GAN). Conditional GAN can be used for same-domain synthesis [10]–[12], such as synthesizing X-ray images with desired abnormalities. CycleGAN can be applied for synthesizing across imaging modalities [14], [15], [35], such as from CT to MRI. GAN based methods are also employed for synthesizing tumors [16], [17], but pixel-wise tumor masks are required because they need to learn pathological patterns from the annotated tumor areas. In comparison, some works propose to synthesize tumors based on morphological operations without requiring any labeled tumor images [18], [19].

B. Tackling Domain Shift

Directly applying the model trained on the synthetic images to the real images performs poorly due to domain shift, and tackling the domain shift problem has been extensively studied. As a promising solution, unsupervised domain adaptation has attracted considerable attention since labels of the target domain are usually unavailable. These methods attempt to bridge the domain gap with different ideas, such as image/feature alignment [20]–[24] or shared proxy tasks [25]–[28], and have achieved impressive adaptation performance. However, most of these approaches require that source data and target data should be available at the same time, which may not always be feasible in reality. An alternative approach is to directly adapt the model on the test data from the target domain [29]–[32], which can address the domain shift problem when source data is not available during test-time. Our method falls in this line of research, and attempts to tackle the synthetic-to-real domain shift problem in the liver tumor segmentation task.

C. Adaptation on Test Sample

The idea of directly fine-tuning the model on the test sample with self-supervision is recently proposed in [29], namely Test-Time Training. A self-supervised auxiliary task is added for updating part of the model parameters on the test data, aiming to improve the performance of the main task. After this seminal work, a test-time entropy minimization scheme is proposed in [30]. In the context of medical imaging, some pioneer works have investigated adaptation on the test sample. A shallow image normalization layer is adapted to minimize the denoising autoencoder's reconstruction loss in [31], while a set of adaptors are trained to minimize the domain shift as measured by the autoencoders' reconstruction loss in [32]. However, previous methods need to freeze the majority of overall parameters of the target model such that they might not be able to provide sufficient flexibility to obtain ideal results. In comparison, our method adapts the model to each test sample with self-supervision from the auxiliary task, *i.e.*, two-stage reconstruction, and all its parameters can be updated during adaptation, allowing the model to work at full capacity.

D. Training Technique in RNN

Recurrent neural network (RNN) has great advantages in serialization tasks, but is hard to train. Teacher forcing is a training technique widely used in RNN [34], where the true previous output is passed as input for predicting later outputs, aiming to reduce the training difficulty. However, true previous outputs are not available during inference time and are replaced by predictions generated by the model itself. The discrepancy between training and inference could lead to poor performance due to bias explosion, but it can be improved by scheduled sampling [36], [37]. We note that teacher forcing is seldom studied outside of RNN, and a pioneer work adopts teacher forcing to train networks for MR image analysis [38]. We treat two-step reconstruction as a serialization task, and design a scheduled mixture strategy to reduce its training difficulty inspired by the training technique in RNN.

III. METHOD

In this paper, we propose a novel Synthetic-to-Real Test-Time Training (SR-TTT) method for learning from synthetic images in the liver tumor segmentation task. Instead of simply sharing parameters between the main task and the auxiliary task as in the original TTT method, we propose to increase their correlation with an auxiliary two-step reconstruction task which takes the output from the main task as its input, such that training with self-supervision from the auxiliary task will directly contribute to the main task. Synthetic tumor images can be obtained based on [19], and are used to train the networks of the main task and the auxiliary task. During TTT, only the main segmentation network is fine-tuned on each test image by minimizing the self-supervised reconstruction loss. An illustration of our proposed SR-TTT is shown in Fig. 3.

A. Training

In this subsection, we introduce the training process of our proposed method. Note that we only require healthy training images x_{train} , and inject synthetic tumors to these healthy images to form training pairs: synthetic image x_{syn} and synthetic tumor mask m_{syn} . In this way, only image-level labels are required rather than pixel-level labels.

1) *Model Architecture*: Main task network is a segmentation network for segmenting the tumors in the input pathological image; auxiliary task network 1 is a generation network for transforming an input pathological image to a pseudo-healthy one without tumors; auxiliary task network 2 is a reconstruction network for reconstructing the input pathological image by combining the healthy image with the tumor segmentation mask. Below we elaborate each network in SR-TTT.

a) *Segmentation network*: The segmentation network segments the tumors in the synthetic pathological image x_{syn} , and outputs a segmentation map $\tilde{m}_{syn} = S(x_{syn})$. The network follows a U-Net architecture, and it is trained with the cross entropy loss defined as follows:

$$\mathcal{L}_{seg}^p = - \sum_i m_{syn}^i \cdot \log(S(x_{syn}^i)), \quad (1)$$

where $S(\cdot)$ is the output from the segmentation network, and m_{syn} is the synthetic groundtruth tumor mask.

Moreover, a healthy image x_{train} has no tumors and its segmentation result should be a black mask with all zero values. We also use healthy images for training the segmentation network to reduce false positives. It is represented by:

$$\mathcal{L}_{seg}^h = - \sum_i m_{black}^i \cdot \log(S(x_{train}^i)), \quad (2)$$

where m_{black} is a black mask with all zero values indicating no tumors in the image.

b) *Generation network*: The generation network aims at transforming the pathological image x_{syn} to a pseudo-healthy one $\tilde{x}_g = G(x_{syn})$, and it also follows a U-Net architecture. Since we obtain the pathological image by injecting synthesized tumors into the healthy training images, those healthy images can be directly used as ground truths for

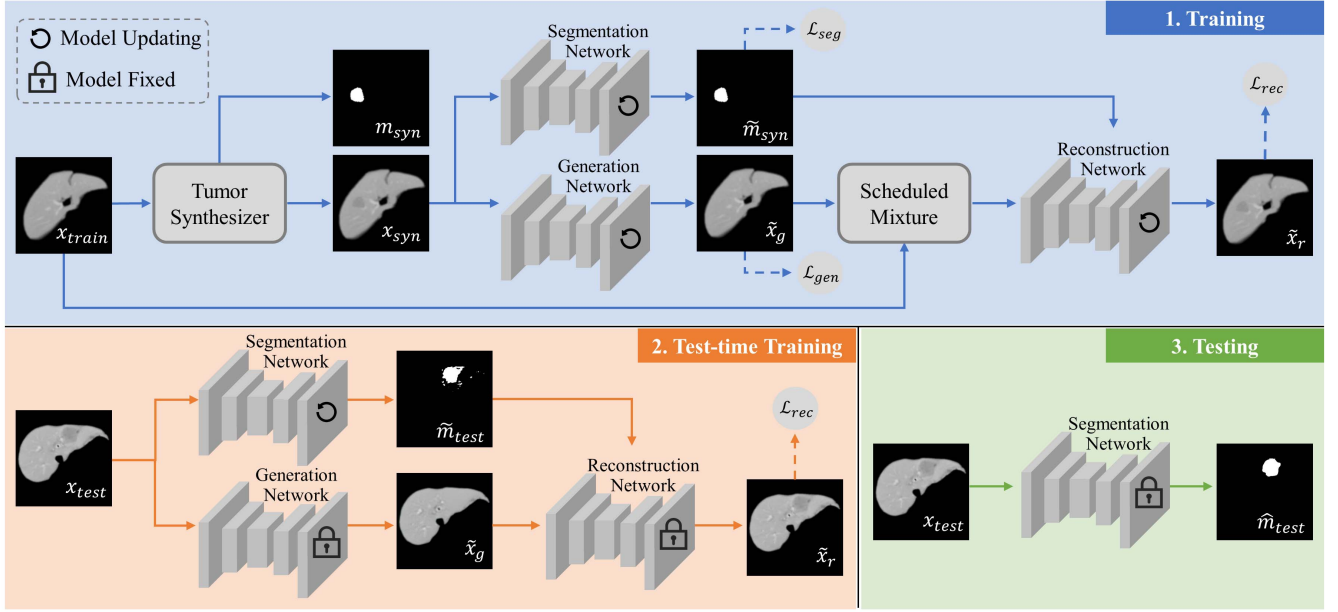


Fig. 3. Overview of the proposed Synthetic-to-Real Test-Time Training (SR-TTT) method for learning from synthetic images in the liver tumor segmentation task. To remove irrelevant information, we only keep the liver regions in the pre-processing step. In the training phase, an auxiliary two-step reconstruction task is added and a scheduled mixture strategy is proposed to train the model. During test-time training, the parameters in the generation and reconstruction models are fixed, and only the segmentation network is updated to adapt to each test image with self-supervision from the auxiliary task. In the testing phase, we can obtain the final predicted liver tumor mask with the updated segmentation model.

supervising the generation network, which is defined as follows:

$$\mathcal{L}_{gen}^g = \|G(x_{syn}) - x_{train}\|_1, \quad (3)$$

where $G(\cdot)$ is the output from the generation network.

Since tumors normally only occupy a small proportion of the whole CT image, simply using the $\mathcal{L}_1$ loss may not achieve satisfactory results for completely removing the tumors. Inspired by [39], the generated pseudo healthy images should not contain tumors such that its segmentation result should also be a black mask with all zero values, which is described as follows:

$$\mathcal{L}_{gen}^s = - \sum_i m_{black}^i \cdot \log(S(G(x_{syn}^i))), \quad (4)$$

where $S(G(\cdot))$ denotes the generated pseudo healthy image from the generation network is then sent to the segmentation network to get its tumor mask, and it is expected to be a black mask with all zero values.

c) *Reconstruction network*: The reconstruction network takes a pseudo healthy image and a tumor segmentation mask as the input, and combines them together to reconstruct a pathological image $\tilde{x}_r = R(\tilde{x}_g, \tilde{m}_{syn})$. Its architecture is the same as the segmentation network and the generation network, but it takes a two-channel input since the two inputs are concatenated. Therefore, the reconstruction network is trained with the loss given by:

$$\mathcal{L}_{rec} = \|R(G(x_{syn}), S(x_{syn})) - x_{syn}\|_1, \quad (5)$$

where $R(G(\cdot), S(\cdot))$ denotes that output from the generation network and the segmentation network are concatenated as input and sent to the reconstruction network.

d) *Model training*: Our proposed auxiliary task is a serialization task, and it is trained together with the main segmentation task in two steps.

In step 1, we first train the generation network, and the total loss in step 1 is denoted as follows:

$$\mathcal{L}_{S1} = \mathcal{L}_{gen}^g + \lambda_{gs} \mathcal{L}_{gen}^s, \quad (6)$$

where λ_{gs} is the balanced weight for the loss items.

In step 2, we train the segmentation network and the reconstruction network. To avoid trivial solutions, we stop the gradients in the generation network in this step. Therefore, the total loss in step 2 is denoted as follows:

$$\mathcal{L}_{S2} = \lambda_{sp} \mathcal{L}_{seg}^p + \lambda_{sh} \mathcal{L}_{seg}^h + \mathcal{L}_{rec}, \quad (7)$$

where λ_{sp} and λ_{sh} are coefficients for the loss items.

2) *Scheduled Mixture*: We treat two-step reconstruction as a serialization task with two steps, which begins from the generation step to the reconstruction step. In the default **free running** mode, the output $\tilde{x}_g$ from the generation step is taken as input of the next reconstruction step. However, the limitation of this default mode is that errors will accumulate due to the wrong predictions from the previous step, and make the model difficult to train. In the **teacher forcing** mode, ground truth x_g of the generation step is used as the input, *i.e.*, x_{train} is sent to the reconstruction step in our case. **Scheduled sampling** is an improved technique by randomly selecting $\tilde{x}_g$ or x_g during training, which aims to avoid the bias explosion problem in the original teacher forcing scheme.

Considering that text tokens are discrete while image data is continuous, we propose a **Scheduled Mixture** strategy, where $\tilde{x}_g$ and x_g are mixed with a random ratio instead of randomly sampling. Comparison of different training schemes is shown

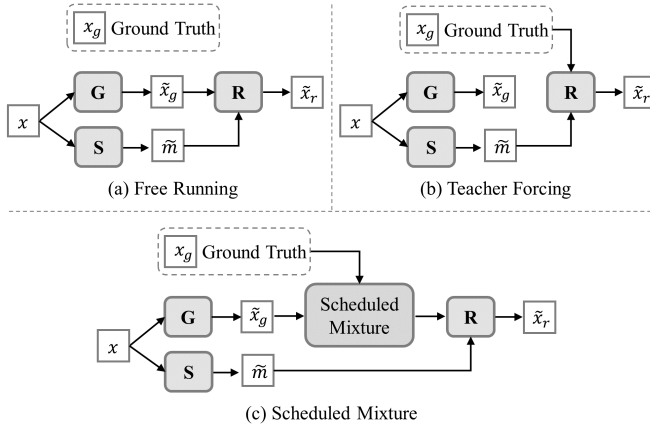


Fig. 4. Comparison of different training schemes.

in Fig. 4. In a nutshell, we mix $\tilde{x}_g$ and x_g in the scheduled mixture strategy as follows:

$$x_{mix} = \lambda x_g + (1 - \lambda) \tilde{x}_g, \quad (8)$$

where λ is a random ratio, $\lambda \in [0, 1]$.

When $\lambda = 1$, the model is trained exactly as the original teacher forcing mode, while when $\lambda = 0$ the model is trained in the same setting as the default free running mode and also as inference. We propose to use a schedule to decrease λ gradually based on a linear decay function as follows:

$$\lambda = \max(0, 1 - c \cdot e_i), \quad (9)$$

where c is the coefficient in the linear decay function and e_i is the training epoch.

B. Test-Time Training

The models trained on the synthetic images can not generalize well to real test images due to the domain shift problem. To address this issue, we further utilize test-time training to adapt the trained model to each test image by directly fine-tuning the model on the test image with self-supervision from the auxiliary task.

Given a test image x_{test} , we first freeze the parameters in the generation and reconstruction networks, and only update the segmentation network with the reconstruction loss. The output from the segmentation network is taken as the input of the reconstruction network and all other networks are frozen, therefore only an improved segmentation result can reduce the reconstruction loss. Ideally, the updated segmentation network can produce a better tumor mask and thus minimize the reconstruction loss, which is defined as:

$$\mathcal{L}_{rec} = \|(R(G(x_{test}), S^*(x_{test})) - x_{test})\|_1, \quad (10)$$

where S^* is the updated segmentation network, while generation network G reconstruction network R remain fixed at test time. The loss of the auxiliary task only requires the test image itself without requiring any manual labels. The entire test-time training procedure is illustrated in Algorithm 1.

Algorithm 1 Test-Time Training

Input: θ_s : parameters of segmentation network S ,

θ_g : parameters of generation network G ,

θ_r : parameters of reconstruction network R ,

x_{test} : test sample,

α : learning rate,

N_{update} : number of update steps

Output: θ_s^* : parameters of updated segmentation network S^*

Initialize θ_s^* with θ_s

for $step = 1, 2, \dots, N_{update}$ **do**

Calculate auxiliary reconstruction loss:

$$\mathcal{L}_{rec} = \|(R(G(x_{test}), S^*(x_{test})) - x_{test})\|_1$$

Optimize segmentation model parameters:

$$\theta_s^* \leftarrow \theta_s^* - \alpha \nabla \mathcal{L}_{rec}$$

end for

return θ_s^*

C. Testing

After test-time training, we can get the updated segmentation model which is adapted to each specific test image. In our proposed SR-TTT, all parameters of the segmentation network can be updated such that it could provide sufficient flexibility to obtain satisfactory results, which is different from most previous TTT methods where the majority of model parameters are fixed.

In the testing phase, the tumor segmentation map is obtained with the updated segmentation model for each test image, and is used to get the final tumor mask after binarization, which is described as follows:

$$m_{test}^i = \begin{cases} 1, & \text{if } S^*(x_{test}^i) > \tau; \\ 0, & \text{otherwise,} \end{cases} \quad (11)$$

where τ is the threshold for binarization.

IV. EXPERIMENTS

A. Datasets and Experimental Settings

1) **Datasets:** We evaluate our method using two public datasets for liver tumor segmentation:

- Medical Segmentation Decathlon Task08 (MSD08) dataset is collected by Memorial Sloan Kettering Cancer Center in USA [40]. This dataset contains portal venous phase CT volumes with a variety of primary and metastatic liver tumours. We follow the data split setting in [19], where 161 CT volumes are used for training and 142 CT volumes for testing. Note that we only use those CT slices with liver but without tumors for training, therefore 4,990 healthy liver images are used for training in MSD08 dataset.
- The Liver Tumor Segmentation Benchmark (LiTS) dataset is collected by several clinic sites from different countries. This dataset contains 201 contrast-enhanced CT volumes from patients with liver diseases. We follow the official data split setting in [41], where 131 CT volumes are used training and 70 CT volumes are used for training. Similar to MSD08, 11,841 CT slices are selected as healthy images for training in LiTS dataset.

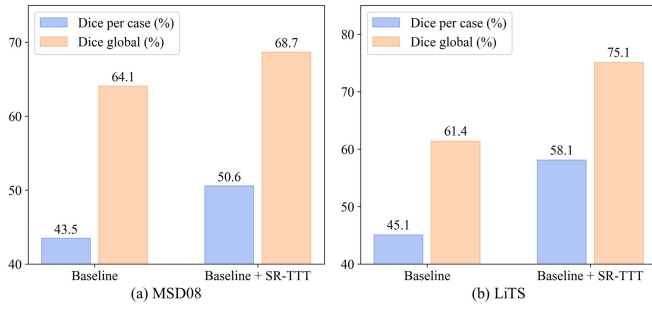


Fig. 5. Effectiveness of SR-TTT. **Baseline:** directly applying the model trained using synthetic images to the real test images; **Baseline + SR-TTT:** adding our proposed SR-TTT to solve the synthetic-to-real domain shift problem for improving the performance of tumor segmentation.

2) Evaluation Metrics: We evaluate the predicted segmentation results by comparing them with ground truths using the Dice per case score and Dice global score, following the LiTS challenge.¹ Dice per case is a dice score applied to each case and averaged over all cases, which is affected more by prediction errors in cases with fewer lesions. Dice global is applied across all cases by combining all volumes into one, which is affected more by large tumors.

3) Pre-Processing: During pre-processing, the image intensity values of all CT slices are first truncated to the range $[-200, +250]$, and then normalized to $[0, 255]$. The liver regions can be extracted by a leading liver segmentation model provided by nnU-Net [42],² and the irrelevant details outside the liver regions are ignored for reducing computational cost. We consider a CT slice as ‘healthy’ if it contains liver regions and its corresponding tumor mask is a black mask, *i.e.*, tumors do not exist in this slice. We inject synthesized tumors into these healthy images to train the model based on [19], but we remove their Couinaud segment restrictions. There are two main steps in [19] for synthesizing liver tumors: Superpixel-guided mask generation and Distribution-based intensity generation. In the first step, various tumor masks are generated based on superpixels, and the generated masks can have diverse locations, sizes and shapes based on parameter selection. In the second step, intensity of each pixel in the mask area is sampled from a distribution function, and the synthetic tumors can appear hyperdense or hypodense, homogeneous or heterogeneous with different parameters.

4) Implementation Details: The segmentation network, generation network and reconstruction network all follow a U-Net architecture, but all the batch normalization layers are removed in our method since we find they could degrade the performance due to the small batch size. The input images are resized to 512×512 . During training, we use the Adam optimizer to train all networks for 8 epochs with a batch size of 4, and set the learning rate to $1e-4$. The coefficients for the loss functions are set to $\lambda_{gs} = 0.1$, $\lambda_{sp} = 2$, $\lambda_{sh} = 1$ empirically. The coefficient in the linear decay function is set to $c = 1/2$ for MSD08 dataset, and $c = 1/6$ for LiTS dataset. The binarization threshold is set to $\tau = 0.5$ for

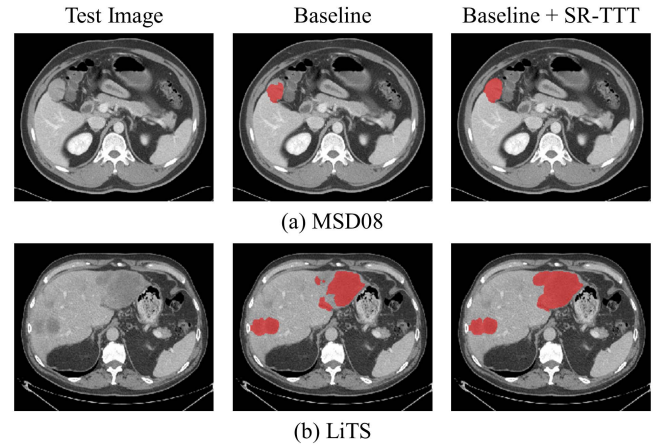


Fig. 6. Examples of liver tumor segmentation results with and without SR-TTT. **Baseline:** directly applying the model trained using synthetic images to the real test images; **Baseline + SR-TTT:** adding our proposed SR-TTT to solve the synthetic-to-real domain shift problem for improving the performance of tumor segmentation.

MSD08 dataset, and $\tau = 0.6$ for LiTS dataset. During test-time training, we set the learning rate to $1e-5$ and perform 5 gradient updates by default. The batch size is set to 1 since the segmentation network is adapted for each test image. All the experiments are conducted on NVIDIA A100 GPUs. It takes around 4 hours for training on MSD08 dataset, and 6 hours for LiTS dataset. During test-time training, it takes around 0.8s for adaptation on each test image. Code of our method is publicly available.³

B. Self Evaluation

1) Effectiveness of SR-TTT: In this subsection, we conduct experiments to analyze the effectiveness of our proposed SR-TTT, as shown in Fig. 5. Directly applying the segmentation model trained using synthetic images on the real test images is the baseline for comparison in our experiment (termed as **Baseline**). SR-TTT adapts the trained model to each test image for improving its segmentation result (termed as **Baseline + SR-TTT**). We observe that the tumor segmentation performance has an obvious improvement after adding SR-TTT. For MSD08 dataset, the dice per case score is increased from 43.5% to 50.6% and the dice global score is increased from 64.1% to 68.7%. The improvement is more obvious for LiTS dataset, where the dice per case score is increased by 13.0% and the dice global score is increased by 13.7%. LiTS dataset contains contrast-enhanced CT volumes which have more complex textures than the portal phase CT volumes in the MSD08 dataset, since structures are highlighted in contrast-enhanced CT volumes to help obtain functional information about tissues. The synthesized tumors normally show simple patterns and the domain shift problem is more serious in the LiTS dataset, therefore reducing the synthetic-to-real domain gap could bring more benefit. Fig. 6 shows some liver tumor segmentation results with and without SR-TTT for MSD08 and LiTS. After adding our proposed SR-TTT, the

¹<https://competitions.codalab.org/competitions/17094>

²<https://zenodo.org/record/4003545>

³<https://github.com/FeiLyu/SR-TTT>

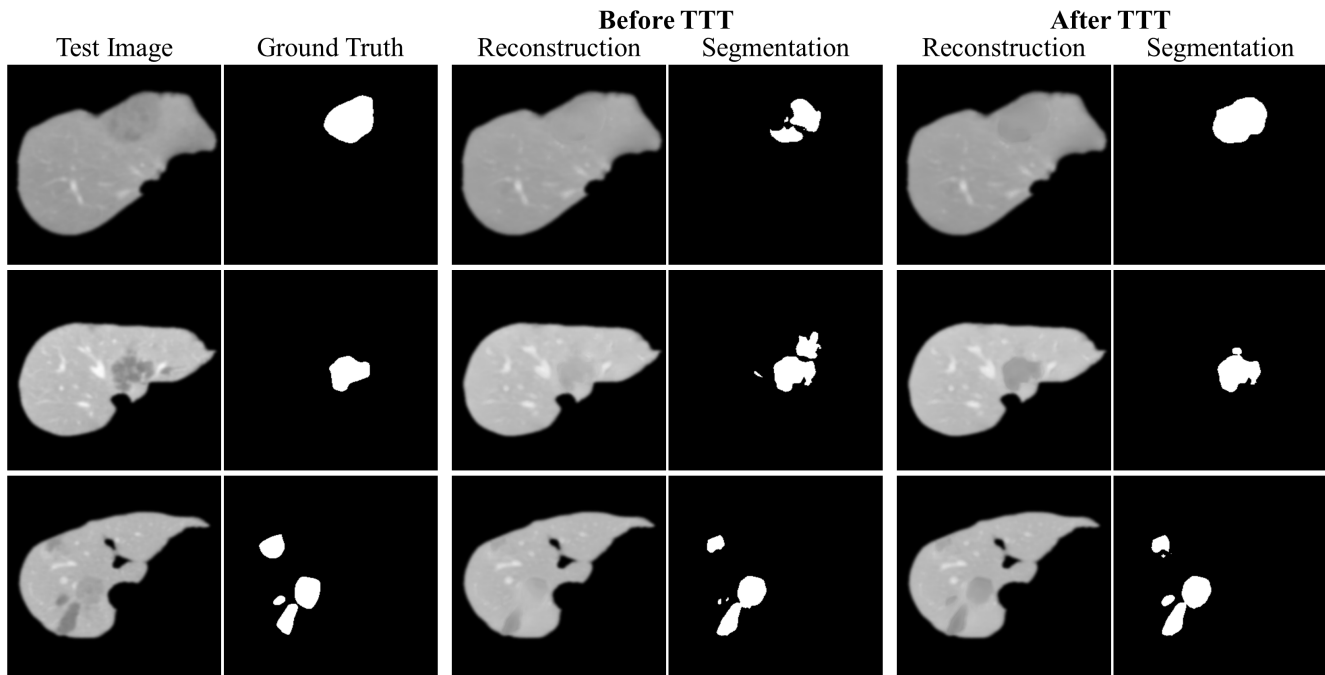


Fig. 7. Visualization results of the main segmentation task and the auxiliary reconstruction task. Before TTT, the model trained using synthetic images is directly applied on the real test images, the segmentation performance is poor and its reconstruction result is also not good. After TTT, the segmentation network is fine-tuned on each test image by minimizing the self-supervised reconstruction loss, the reconstruction result is more visually pleasing and the segmentation performance is also significantly improved.

tumor segmentation results are more complete. The result verifies that solving the domain shift problem is quite important for learning from synthetic images, which further demonstrates the effectiveness of our proposed SR-TTT.

2) Analysis on Two-Step Reconstruction: SR-TTT requires a self-supervised auxiliary task, *i.e.*, two-step reconstruction, to help the main task adapt to each test image at test time. To better understand the correlation between the main segmentation task and the auxiliary reconstruction task, we provide visualization results of these two tasks before and after TTT in Fig. 7. When directly applying the model trained using the synthetic images on the real test images, the segmentation result is poor due to the domain shift problem. Since the segmentation result is taken as the input and fed to the reconstruction network, the reconstruction result is also not satisfactory. After TTT, the segmentation network is updated to adapt to each test image by minimizing the self-supervised reconstruction loss, the reconstructed images are more visually pleasing. Since the parameters of the generation network and the reconstruction network are fixed, only improved tumor segmentation outputs can lead to better reconstruction results, which is the explicit correlation we would like to build between these two tasks in the SR-TTT method. To conclude, the visualization results show that the auxiliary reconstruction task is explicitly related to the main segmentation task, and training with self-supervision from the auxiliary task at test time will directly contribute to the segmentation performance of the main task on each test image.

3) Analysis on Scheduled Mixture: This part evaluates different strategies for training the networks in the proposed

TABLE I
COMPARISON OF LIVER TUMOR SEGMENTATION RESULTS
USING DIFFERENT TRAINING STRATEGIES

Training Strategy	Dice per case(%)	Dice global(%)
Free Running	45.4	67.6
Teacher Forcing	48.8	68.5
Uniformed Sampling	47.3	63.2
Scheduled Sampling	46.7	66.1
Uniformed Mixture	45.4	67.2
Scheduled Mixture	50.6	68.7

SR-TTT. Besides the **free running** mode and **teacher forcing** mode, we also train the model with a **scheduled sampling** strategy, which degrades the sampling probability from ground truths with the same linear decay function as our schedule mixture strategy. **Uniformed sampling** indicates the sampling probability stays 50% while **uniformed mixture** indicates the mixture ratio is always 50%. The results on MSD08 dataset using different training strategies are shown in Table I. Experiment results in Table I demonstrate that our proposed **schedule mixture** strategy performs better than the other training strategies. The main reason is that schedule mixture strategy can avoid the error accumulation problem in the free running mode and also mitigate the bias explosion caused by the original teacher forcing strategy. Moreover, the original scheduled sampling is designed for text sequence prediction tasks where text tokens are discrete. In our task, image data is continuous and mixing two images is a commonly used method in computer vision tasks rather than sampling. At the beginning of training, the predicted output of the generation model itself is not good since the generation model is not

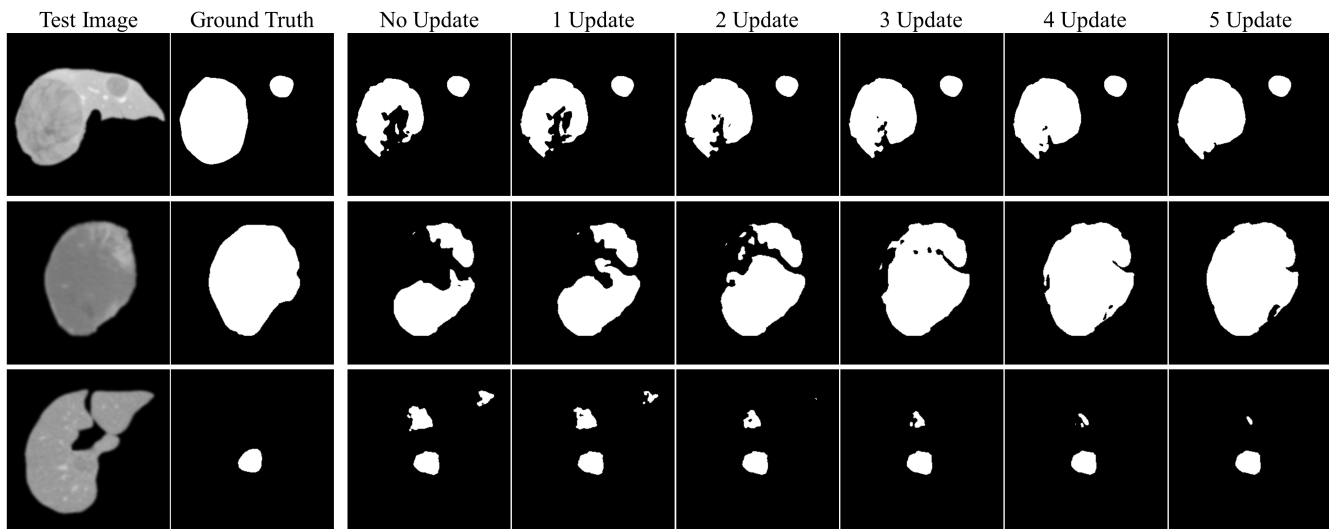


Fig. 8. Visual illustration of the unfolded test-time training process. The segmentation results are gradually improved with more gradient updates and are visually closer to the ground truths.

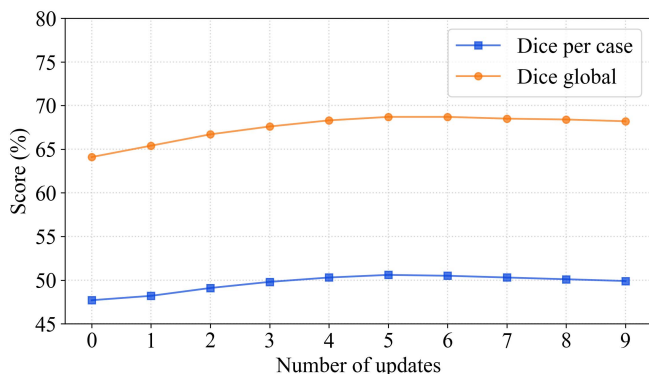


Fig. 9. Evaluation of the number of gradient updates for test-time training on MSD08 dataset.

TABLE II

COMPARISON OF LIVER TUMOR SEGMENTATION RESULTS USING DIFFERENT UPDATING SCHEMES

Updating Scheme	Dice per case(%)	Dice global(%)
S	50.6	68.7
S+G	50.5	68.3
S+R	50.2	68.4
S+G+R	50.3	68.1

well trained, which could lead to very slow convergence, so a higher mixture ratio from the ground truth should help; at the end of training, a higher mixture ratio from predicted output is preferable since this corresponds to the true inference situation. In this way, the scheduled mixture strategy can avoid error accumulation and also reduce the discrepancy between training and inference, leading to better training performance.

4) *Analysis on Test-Time Training*: Our method attempts to solve the synthetic-to-real domain shift problem by fine-tuning the model on each test image. Number of gradient updates is an important factor in our method, and we first show its impacts on the segmentation result in Fig. 9. We observe that more updates tend to produce better results in the initial

stage, but the performance starts to drop when we update the model 5 to 6 iterations. In our experiments, we perform 5 gradient updates by default, since it usually achieves decent performance in most cases. To gain further insights of our method, we unfold the test-time training process and show the segmentation result after each gradient update in Fig. 8. When there is no update, the model trained on the synthetic images is directly applied to test images, and we can find that the segmentation performance is poor due to the domain shift problem. Even the network is only updated once, we can see obvious improvement in the segmentation results. With more updates, the segmentation results are gradually improved and are visually closer to the ground truths. The segmented tumor masks of test samples in the first and second rows are very incomplete initially and gradually become complete with more gradient updates, while false positives of the test sample in the third column are obvious with no updates and are gradually removed to get better results. Moreover, we also conduct experiments where the other two networks are also updated during test-time training for comparison. The results in Table II demonstrate that only updating the segmentation network could get the optimal performance. The reason is that updating the other two networks is more biased to only improve the reconstruction quality, and the contribution from the segmentation network is weakened. To conclude, we empirically demonstrate that the test-time training process in our method can successfully adapt the model to each test image and achieve better performance of liver tumor segmentation.

C. Comparison With State-of-the-Arts

1) *Competing Methods for Tackling Domain Shift*: Test-time training is the key process for tackling the synthetic-to-real domain shift problem in our SR-TTT method. In this part, we compare our SR-TTT method with leading methods for tackling domain shift, including the original self-supervised test-time training [29], Autoencoder (AE) based

TABLE III

COMPARISON OF DIFFERENT METHODS FOR TACKLING SYNTHETIC-TO-REAL DOMAIN SHIFT IN LIVER TUMOR SEGMENTATION

Method	Auxiliary Task/Module	MSD08		LiTS	
		Dice per case(%)	Dice global(%)	Dice per case(%)	Dice global(%)
Baseline	×	43.5	64.1	45.1	61.4
Joint-Training	Rotation [29]	29.9	39.4	30.8	69.4
	Reconstruction	45.7	66.3	54.7	69.0
AE-TTA	Autoencoder [32]	47.6	65.5	53.6	68.5
DAE-TTA	Denoising Autoencoder [31]	45.8	65.4	52.4	69.6
UDA-SS	Rotation [25]	32.2	45.3	36.6	72.5
	Reconstruction	44.7	61.6	48.9	65.6
Ours	Two-stage Reconstruction	50.6	68.7	58.1	75.1

TABLE IV

COMPARISON OF DIFFERENT METHODS FOR LIVER TUMOR SEGMENTATION

Method	Supervision	MSD08		LiTS	
		Dice per case(%)	Dice global(%)	Dice per case(%)	Dice global(%)
nnU-Net [42]	Pixel-level label	70.5	86.7	76.2	84.5
CouinaudNet [19]	Couinaud segment label	62.2	74.0	68.4	80.9
VAE [43]	Image-level label	30.4	50.1	43.0	65.9
AAE [44]	Image-level label	33.9	53.7	48.5	69.2
f-AnoGAN [45]	Image-level label	31.1	54.4	41.2	66.1
Ours	Image-level label	50.6	68.7	58.1	75.1

test-time adaptation [32], Denoising Autoencoder (DAE) driven test-time adaptation [31], and unsupervised domain adaptation through self-supervision [25]. These methods are compared on the liver tumor segmentation task under the same setting, and the results are shown in Table III.

The baseline method (termed as **Baseline**) directly applies the model trained on synthetic images to the test images without adaptation. Sharing the same feature backbone is the key idea in [29] for adaptation on test images. Based on their idea, we replace the shared encoder network in [29] with our U-Net encoder network, and apply it to our liver tumor segmentation task (termed as **Joint-Training**). Rotation prediction is used as the auxiliary task in [29] for updating the shared encoder which works well for the classification task, but we empirically find that its performance is disappointing in our task. One possible reason is that CT images have a standard structure based on human anatomy which makes it easy for the model to predict its rotation. The model tends to learn some landmarks, such as vertebra, rather than the tumors, which is likely to hurt the performance of liver tumor segmentation. Following the spirit in [29], we replace rotation prediction with image reconstruction, and the segmentation performance is significantly improved. This demonstrates that the idea of test-time training proposed in [29] is effective only when a special-purpose auxiliary task is well designed. [32] and [31] are two recent works which also aim to tackle the domain shift problem in medical image segmentation. We also replace their networks with our U-Net, but follow their ideas for adaptation on test images (termed as **AE-TTA** and **DAE-TTA**, respectively). The results show these two methods can mitigate the domain shift problem and improve the segmentation result after adaptation on each test sample. However, they are originally designed to deal with the domain shift due to global appearance difference, such as modality change or scanner change, which is not the case

in our problem since we only change some local areas of healthy images with synthetic tumors. Moreover, the DAE modeled as a 3D-CNN is a key component for driving the test-time adaptation in [31], but we replace it with a 2D-CNN since our model takes 2D image as input, which may also weaken its power. We also conduct experiments using a recent unsupervised domain adaptation method (termed as **UDA-SS**). UDA-SS aims to find an invariant representation for both training distributions and test distributions by learning to perform a shared self-supervised task. We get the similar results as in Joint-Training, using reconstruction preforms much better than rotation prediction. Therefore, how to choose an auxiliary task is non-trivial for both Joint-training and UDA-SS. Moreover, even though test-time adaptation methods based on entropy minimization have achieved success in many applications [30], they are not applicable in our model since we have removed all batch normalization layers due to small batch size.

To conclude, results in Table III demonstrate that our SR-TTT achieves the best performance among all the competing methods for tackling the synthetic-to-real domain shift problem in the liver tumor segmentation task. Competing methods designed for image classification usually achieve disappointing results for the tumor segmentation task, and specific adjustments are required. Competing methods designed for medical image segmentation mainly focus on the domain shift due to global appearance difference, while the synthetic-to-real domain shift is due to local area changes in our problem.

2) Competing Methods for Liver Tumor Segmentation: In Table IV, we first evaluate the liver tumor segmentation performance by using different levels of supervision, including pixel-level label, Couinaud segment label (0: healthy, 1 ~ 8: tumor location), and image-level labels (0: healthy, 1: pathological), and the difficulty of collecting labels is successively reduced. nnU-Net [42] trained with pixel-level labels can be regarded as the upper bound for each dataset.

Weakly supervised methods using Couinaud segment labels shows degraded performance for liver tumor segmentation. Image-level labels are much weaker than these two types of labels, but collecting image-level labels is much easier and cheaper. Moreover, our method only requires healthy training images and utilizes infinite synthetic data for training. Despite the performance gap between our method and those methods with stronger supervision, our method can be seen as an encouraging step for utilizing synthetic data in medical image segmentation tasks, which performs decently but requires significantly less manual annotation effort.

We also compare our method with existing deep learning based methods for anomaly segmentation when only image-level labels are given [46]. These leading methods, including VAE [43], AAE [44], **f-AnoGAN** [45], have shown promising performance in brain MR images. These methods can be adapted for liver segmentation, where they use healthy training images to train generative models, then tumor masks of pathological training images are inferred with the trained model, finally a fully supervised segmentation model is trained with all training images and their pseudo tumor masks. Unlike these methods, our proposed SR-TTT only requires healthy training images and inject synthesized tumors to these healthy images for training. Because there are still differences between synthetic and real tumors, the segmentation model trained using synthetic tumor images shows degraded performance on real tumor images due to domain shift. Our method can properly handle the synthetic-to-real domain shift problem, so as not to degrade the practical value of the synthetic data. The results shown in Table IV demonstrate that our method shows obvious superiority to other competing methods when only image-level labels are available. In our method, the liver tumor segmentation model is trained with almost infinite image samples from the synthetic data, and it is fine-tuned on each test image to avoid the negative impacts caused by the domain shift problem. As a result, our model trained on synthetic images can generalize well on real test images.

V. DISCUSSION

Though perfectly labeled synthetic data can be acquired with low cost, directly applying the model trained using the synthetic data on real test samples performs poorly due to the domain shift problem. In this paper, we propose a method based on test-time training to bridge the domain gap between real and synthetic CT images for liver tumor segmentation. The experiment results on two public datasets have demonstrated the effectiveness of our proposed method.

Compared to conventional deployment of deep networks, the cost of our method is a short adaptation time for each test image (about 0.8s in our experiments) and additional network parameters (around 60M). Our paper is not as concerned about computational efficiency as improving the segmentation performance, here we provide two potential solutions that might help improve the efficiency, but have not been thoroughly verified. The first one is to improve the network design. Currently, 3 networks all adopt the vanilla U-Net architecture in our method, and utilizing advanced light-weight network architectures could help reduce the computation cost.

The second is to explore new adaptation schemes. Currently, test-time training is performed image-by-image, and we could explore how to fine-tune the model on the test samples batch-by-batch or volume-by-volume, so as to improve the efficiency.

Another key that should be explored in the future study is how to design a better tumor synthesizer. The performance of tumor synthesizer will not directly affect the proposed SR-TTT, but will directly affect the tumor segmentation performance. If the tumor synthesizer is perfect and can synthesize tumors with the same distribution as real tumors, then the domain shift problem does not exist and all techniques for reducing domain gap are not necessary. Therefore, a better tumor synthesizer and our proposed method are two ways for improving the tumor segmentation performance. If we can develop a more generalizable tumor synthesizer, our method can be extended to other organs, such as brain [47], lung [18] or pancreas [48]. To adapt to different organs, incorporating prior knowledge in clinical practice into the method would be very helpful. We will try to explore more sophisticated designs of tumor synthesizers and plan to leave it for future work.

VI. CONCLUSION

In this paper, we propose a novel approach, namely Synthetic-to-Real Test-Time Training (SR-TTT), for improving the performance of liver tumor segmentation models trained using synthetic images. We update the model parameters for each test image with self-supervision from an auxiliary two-step reconstruction task, and introduce a scheduled mixture strategy to train the model. Extensive experiments on two public datasets have verified the effectiveness of our propose SR-TTT. Addressing the synthetic-to-real domain shift problem is important in many real-world situations when high-quality labeled data is difficult to collect especially for medical data, whereas infinite synthetic data can be used with no need of human labeling. Our method provides an inspiring solution to realize the full potential of the synthetic data, and could be potentially applied in other applications.

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